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B.Tech. Degree VI Semester Examination April 2019

ME 15-1603 OPERATIONS MANAGEMENT (2015 Scheme)

Time: 3 Hours

Maximum Marks: 60

PART A (Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Define the term operations management.
 (b) What is network crashing?
 (c) What is Dummy activity? Why it is needed in network?
 (d) What is Delphi method of forecasting?
 (e) Define the term standardization.
 (f) What do you mean by expediting in production control process?
 (g) What is a Gantt chart?
 (h) Define the term scheduling in operations.
 (i) What do you mean by predictive maintenance?
 (j) What are the reasons for machine replacement?

PART B

(4 × 10 = 40)

- II. (a) What are the objectives of operations management? (3)
 (b) Draw the network diagram for the project, the details of the project are given in the table. Find out the project duration, critical path and float for each activity. (7)

Activity	Predecessor activity	Duration (Days)
A	-	5
B	-	7
C	-	4
D	A	7
E	C	6
F	A	5
G	B, D, E	8
H	C	6

OR

- III. (a) Discuss moving average method and exponential smoothing methods of forecasting. (5)
 (b) Draw the network diagram from the following information, Activity A is the starting activity. Activity B follows A; C and D follows B; E follows D; F and G both follow C; H follows F, I follows G and E, J follows H, K follows I and L follows K and J. (5)

(P.T.O.)

- IV. (a) Explain the various steps involved in production planning and control. (5)
 (b) How inventories are classified? What are the major costs involved in inventory control process? (5)

OR

- V. (a) What are the essential stages of product design process? Explain with an example of your choice. (5)
 (b) IFFCO produces fertilizer to sell to wholesalers. One raw material calcium nitrate is purchased from a nearby supplier at ₹22.50 per ton. Company estimates that it will need 5,750,000 tons of calcium nitrate next year. The annual carrying cost of this material is 40% of the acquisition cost and the ordering cost is ₹595.
 (i) What is the most economical quantity?
 (ii) How many orders will be placed per year?
 (iii) How much time will be elapsed between orders?

- VI. (a) Define the term aggregate planning? What are the pre-required inputs of aggregate planning? (5)
 (b) What are the principles of scheduling? Explain. (5)

OR

- VII. (a) List and explain various rules or methods followed for sequencing of jobs in a machine shop? (5)
 (b) A set of jobs has to process in 3 machines M1, M2 and M3 respectively in order. The processing time required for the jobs in each machine is given below. Find out the optimum sequence, total minimum elapsed time and idle times in M2 and M3 using Johnsons algorithms.

Jobs	M1 (Hrs)	M2 (Hrs)	M3 (Hrs)
A	4	5	7
B	5	2	8
C	5	3	5
D	6	5	6
E	4	2	8
F	6	1	5

- VIII. (a) You are assigned with a project of setting up of an IT park. What are the factors to be considered while selecting a location? Justify your reasons for not selecting other locations. (5)
 (b) Differentiate between line layout, process layout and combination layout with the help of sketches. (5)

OR

- IX. (a) What are the various principles of material handling? Explain. (5)
 (b) What are the various methods of maintenance of machineries? Discuss the merits and demerits of each one. (5)
